

Moritz Grauer

Graphics Programmer · Game Developer · Researcher

CONTACT

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-  Karlsruhe, Germany

SKILLS

LANGUAGES

- C++
- C
- C#
- Java
- Python
- GLSL
- HLSL

GRAPHICS APIS

- Vulkan
- OpenGL

RENDERING

- Path Tracing
- BVH / SAH
- PBR / GGX
- Global Illumination
- NEE / MIS

TOOLS

- Git
- LaTeX
- CPU Profiling
- GPU Profiling

LANGUAGES

- GermanNative
- EnglishFluent

ABOUT

I build rendering systems and research ray tracing, with a focus on GPU acceleration structures and global illumination. Four years as a researcher and teaching associate at KIT, supervising nine theses and teaching courses on photorealistic rendering and computer graphics. Outside of research I participate in game jams and develop my own path-tracing engine.

EXPERIENCE

- Computer Graphics Researcher @ KIT**
Feb 2022 - Apr 2026 · Karlsruhe
Research on ray tracing acceleration structures (BVH with binned SAH), GPU rendering pipelines, and global illumination. Teaching associate for Photorealistic Image Synthesis, Computer Graphics, and Computer Graphics 2. Supervised 9 theses on topics including wide BVH traversal on GPU, mesh shader volume rendering, and neural BVH construction.
- QA Engineer @ Gameforge**
Oct 2020 - Sep 2021 · Karlsruhe
Quality assurance for live online games. Bug tracking, test case documentation, and cross-team coordination for release validation.
- Programming Tutor @ KIT**
Oct 2016 - Jul 2019 · Karlsruhe
Led exercise sessions for undergraduate programming courses. Graded assignments and held office hours.

EDUCATION

- M.Sc. Informatics**
KIT · 2018 - 2021
Focus on computer graphics and high-performance computing. Exchange semester at Chalmers University of Technology, Gothenburg.
- B.Sc. Informatics**
KIT · 2015 - 2018

★ Studienstiftung des deutschen Volkes scholarship

TEACHING

Photorealistic Image Synthesis S'22 · S'23 · W'24/25	Computer Graphics 2 S'24 · S'25
Computer Graphics W'22/23 · W'25/26	Lab Course Supervision W'23/24

SELECTED PROJECTS

- VexEngine** [github ↗](#) [project page ↗](#)
Custom C++ rendering engine with dual OpenGL/Vulkan backends. Progressive CPU and GPU path tracing with hardware ray tracing, Cook-Torrance GGX PBR materials, BVH with binned SAH, NEE and MIS light sampling, post-processing pipeline (tonemapping, bloom), and a full dockable editor UI.
- Necrocure** [itch.io ↗](#)
Mini Jame Gam #55 entry. Alien abduction action minigame built in 58 hours. Handled all programming, VFX, and UI; created with one collaborator handling art and audio.
- Quack 3** [ggj ↗](#)
Global Game Jam 2026 entry. Fast-paced FPS arena shooter developed over 48 hours in a team of three, with all art and audio created on-site during the jam.